

Distributed High-Gain Observer Design for Interconnected Nonlinear System

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- 1 Problem Description
- 2 Error Space Decomposition
- 3 Design Algorithm
- 4 Simulation Result
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Background

- Distributed Observer Design:
- Interconnected System:
- Algorithm Proposition:

Interconnected Nonlinear Systems

We consider the following interconnected system:

$$\begin{cases} \dot{x}_i = A_{n_i} x_i + B_{n_i} \varphi_i(x_1, \dots, x_i, u_i) \\ y_i = C_{n_i} x_i \end{cases}, \quad \forall i \in \mathcal{N}, \quad (1)$$

where

$$A_{n_i} = \begin{bmatrix} 0_{(n_i-1) \times 1} & I_{(n_i-1)} \\ 0 & 0_{1 \times (n_i-1)} \end{bmatrix}, \quad B_{n_i} = \begin{bmatrix} 0_{(n_i-1) \times 1} \\ 1 \end{bmatrix},$$

$$C_{n_i} = \begin{bmatrix} 1 & 0_{1 \times (n_i-1)} \end{bmatrix},$$

with nonlinear function $\varphi_i : \mathbb{R}^{1+\sum_{j=1}^i n_j} \mapsto \mathbb{R}$ satisfying

$$\left\| \varphi_i(x_{[1,i]} + \Delta x_{[1,i]}, u_i) - \varphi_i(x_{[1,i]}, u_i) \right\| \leq L_{\varphi_i} \sum_{j=1}^i \sum_{k=1}^{n_i} |\Delta x_{j,k}|.$$

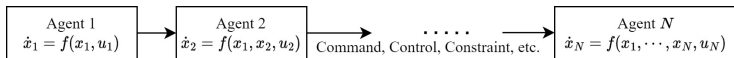


Figure: Interconnected System in Multi-Agents Systems

Distributed Estimation Scheme

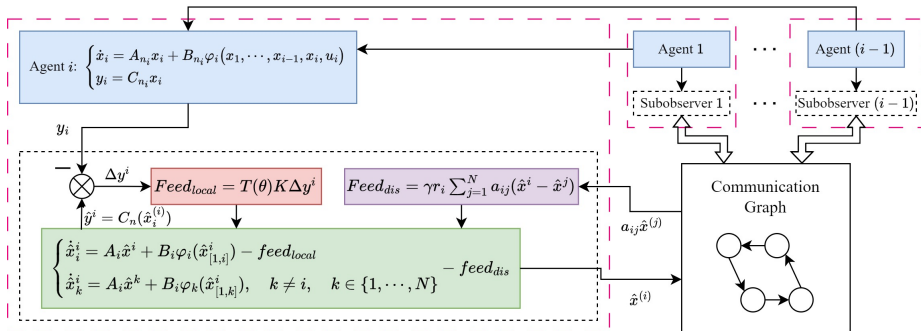


Figure: Distributed Observer Scheme for Interconnected System (1)

Distributed High-Gain Observer

The local high-gain observer implemented at agent i to estimate the collective state x is given by:

$$\begin{aligned} \dot{\hat{x}}^{(i)} = & A\hat{x}^{(i)} + B\varphi(\hat{x}^{(i)}, u) + TL_i(y_i - C_i\hat{x}^{(i)}) \\ & + \gamma TP_i^{-1}T^{-1}r_i \sum_{j=1}^N a_{ij}(\hat{x}^{(j)} - \hat{x}^{(i)}), \quad i \in \mathcal{N}, \end{aligned} \quad (2)$$

where r_i is the i th entry of the positive row vector r . The high-gain matrix T is block-diagonal with $T \triangleq \text{diag}(T_1, \dots, T_N)$, where $T_i \triangleq \text{diag}(\theta_i, \dots, \theta_i^{n_i})$. The feedback matrix, L_i and P_i can be described by

$$\begin{aligned} L_i &\triangleq \left[0_{1 \times \text{sum}(n_{[1, i-1]})}, L_{n_i}^\top, 0_{1 \times \text{sum}(n_{[i+1, N]})} \right]^\top, \\ P_i &\triangleq \text{diag} \left(I_{\text{sum}(n_{[1, i-1]})}, P_{n_i}, I_{\text{sum}(n_{[i+1, N]})} \right). \end{aligned} \quad (3)$$

Theory Scheme

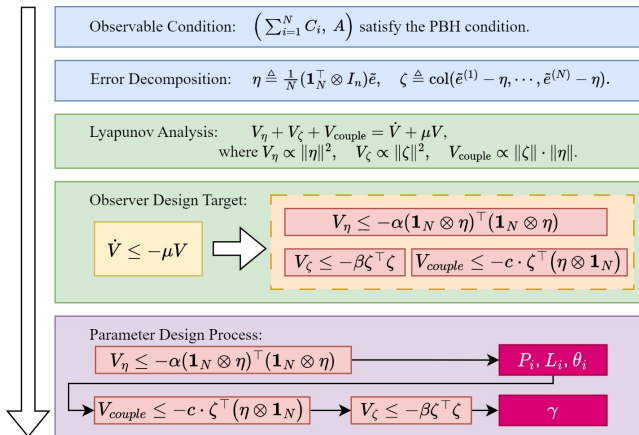


Figure: The main scheme of this work ¹

¹Wu, Isidori, and Lu 2021

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Error Decomposition

Noticed that in the triangular observable form (1), we have not only the observable pair $(\text{row}(C_1, \dots, C_N), A)$, but also $(\sum_{i=1}^N C_i, A)$.

$\implies$ Define the decomposition of the transformed error $\tilde{e}^{(j)}$ of each agent below:

$$\eta \triangleq \frac{1}{N} \sum_{i=1}^N \tilde{e}_i^{(k)} = \frac{1}{N} (\mathbf{1}_N^\top \otimes I_n) \tilde{e}, \quad \zeta \triangleq \text{col}(\tilde{e}^{(1)} - \eta, \dots, \tilde{e}^{(N)} - \eta).$$

And the following inequality can be obtained:

$$\tilde{e}^\top (\hat{\mathcal{L}} \otimes I_n) \tilde{e} = \zeta^\top \left[(U_2 \Lambda_{\hat{\mathcal{L}}} U_2^\top) \otimes I_n \right] \zeta \leq -\gamma \lambda_2 \zeta^\top \zeta,$$

where $\Lambda_{\hat{\mathcal{L}}} = \text{diag}(m, \lambda_2, \dots, \lambda_N)$ and λ_i follow the same definition in Lemma ??, and m is a scalar can be randomly chosed since $\sum_{i=1}^N \zeta^{(i)} \equiv 0$.

Lyapunov Analysis

??? (The lyapunov decomposition and quadratic forms)

Auxiliary Result

Proposition

Let the matrices M_i and M_{ik} be the components of V_η as defined in (15). If the following matrix inequality holds for all $i \in \{1, \dots, N\}$:

$$M_i + N\alpha \cdot I_{n_i} + \sum_{k=1}^{i-1} \|M_{ik}\| I_{n_i} + \sum_{k=i+1}^N \|M_{ki}\| I_{n_i} \leq 0. \quad (4)$$

Then the average error component V_η is upper bounded by $V_\eta \leq -\alpha(\mathbf{1}_N \otimes \eta)^\top (\mathbf{1}_N \otimes \eta)$.

Proposition

Let the matrices $W_i^{(j)}$ and $W_{i,k}^{(j)}$ be the components of V_ζ as defined in (16). If the following matrix inequality holds for all $i, j \in \mathcal{N}$:

$$W_i^{(j)} + \beta I_{n_i} + \sum_{k=1}^{i-1} \|W_{ik}^{(j)}\| I_{n_i} + \sum_{k=i+1}^N \|W_{ki}^{(j)}\| I_{n_i} \leq 0. \quad (5)$$

Then the consensus error component V_ζ is upper bounded by $V_\zeta \leq -\beta \zeta^\top \zeta$.

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Main Theorem

Theorem

Consider the MAS (1) and distributed high-gain observer (2). Let the convergence rate $\mu > 0$ and i th agent's local stability margins $\tau_i > 0$ be given. The estimation error $e^{(i)}$ converges exponentially to zero for all $i \in \mathcal{N}$ if there exist symmetric positive definite matrices P_{n_i} , full matrices R_{n_i} , high-gain parameter $\theta_i > 1$ and a consensus gain $\gamma > 1$ satisfying the following conditions:

$$\text{Sym}(P'_i A_{n_i} - R_i C_{n_i}) \leq -\tau_i P'_i, \quad (6)$$

where $P'_i = (N - 1)I_{n_i} + P_{n_i}$. The high-gain parameters θ_i are chosen to satisfy the following inequality for all $i \in \mathcal{N}$:

$$\theta_i \geq \frac{\mu + N + (\sqrt{n_i} + 1)L_{\varphi i}}{\tau_i} + \sum_{k=1}^{i-1} \frac{\theta_i^{-n_i} \theta_k^{n_k} \sqrt{n_k} \|P'_{n_i}\| L_{\varphi i}}{\tau_i \lambda_{\min}(P'_{n_i})} + \sum_{k=i+1}^N \frac{\theta_k^{-n_k} \theta_i^{n_i} \sqrt{n_i} \|P'_{n_k}\| L_{\varphi k}}{\tau_i \lambda_{\min}(P'_{n_i})}. \quad (7)$$

And the consensus gain γ is chosen such that:

$$\gamma \geq \frac{\beta + Y_{\max}}{\lambda_2}, \quad (8)$$

where $Y_{\max}$ is an upper bound on the coupled error dynamics. The positive scalars β and $Y_{\max}$ are functions of the system and observer parameters that arise from the Lyapunov proof. Their explicit definitions are provided in Appendix ??.

Algorithm

Algorithm 1: Observer Design Procedure

Inputs : $A_{n_i}, C_{n_i}, \mathcal{L}, r, \mu > 0, \tau_i > 0, L_{\varphi i}$ for $i \in \mathcal{N}, \alpha \gg 1, c = 0, Y_{\max} = 0;$

Outputs: $P_{n_i}, L_{n_i}, \theta_i > 1, \gamma > 1;$

Step 1: Calculate feedback matrices:

for $i = 1$ **to** N **do**

 Define $P'_i = (N - 1)I_{n_i} + P_{n_i};$

 Solve the LMI (6) for $P_{n_i} > 0$ and $R_{n_i};$

$L_{n_i} \leftarrow P_{n_i}^{-1} R_{n_i};$

end

Step 2: Calculate high-gain parameters:

for $i = 1$ **to** N **do**

for $k = 1$ **to** $i - 1$ **do**

$\theta_i \leftarrow \max(1, \text{RHS of } (??));$

end

$\theta_i \leftarrow \max(1, \text{RHS of } (??));$

end

Step 3: Calculate the consensus gain:

for $i = 1$ **to** N **do**

for $k = 1$ **to** $i - 1$ **do**

$Y_{\max} \leftarrow \text{RHS of } (??);$

$c \leftarrow \max(1, \text{RHS of } (??));$

end

$\alpha \leftarrow \min\left(\alpha, \lambda_{\min}\left(P_{n_i} + (N - 1)I_{n_i}\right)\right);$

end

$\beta \leftarrow \frac{c^2}{\alpha}; \quad \lambda_2 \leftarrow \min\left\{\lambda\left(R\mathcal{L} + \mathcal{L}^\top R\right) \mid \lambda > 0\right\}; \quad \gamma \leftarrow \max\left(1, \frac{(\beta + Y_{\max})}{\lambda_2}\right).$

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Leader Cruising Control (LCC) Based Vehicle Platoon

??? (Model Description)

Simulation Result

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